Gaillaume Argand

The Atom, Periodic Table & Chemical Formula

Name: Oisín Argand

$\frac{115}{115}$ = 100% SUPER! Well Done Oisín!!

Jan 26

Question 1

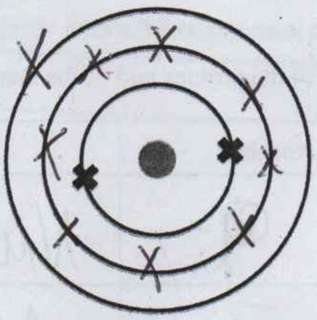
(b) An atom of sodium has an atomic number of 11 and a mass number of 23. (24)

For examiner use only (1) (2)

(i) How many protons does this atom contain? 11 3

(ii) How many neutrons does this atom contain? 12 3

(iii) Complete the diagram below, drawing all the electrons on the Bohr structure for this sodium atom. The two electrons on the first shell are already drawn for you.



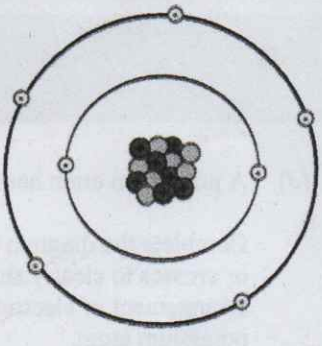
6

(iv) Name the Group of the Periodic Table to which sodium belongs.

Alkali Metals 3

Question 2

(h) The diagram shows an atom that has seven positive sub-atomic particles.



(i) Name the type of sub-atomic particle that has a positive charge.

Proton 3

(ii) Name the type of sub-atomic particle that has a negative charge.

Electron 3

(iii) Name the part of the atom where the positive charges are located.

Nucleus 2

(iv) Identify the element whose atom is shown in the diagram.

Nitrogen 2

Question 3

use only

(24) (1) (2)

(*c*) A student researched the element fluorine (F) and found the following information.

9
F
19

2,7

(*i*) Name one particular source which the student could have used to find this information. Periodic Table

2

(*ii*) Using this information, show in the table below the *number* of each type of sub-atomic particle present in an atom of fluorine. Complete the table to show the *location* (within the atom) of the protons and of the neutrons.

Particle	Number	Location
Proton	9	Nucleus
Neutron	10	Nucleus
Electron	9	Electron cloud

10

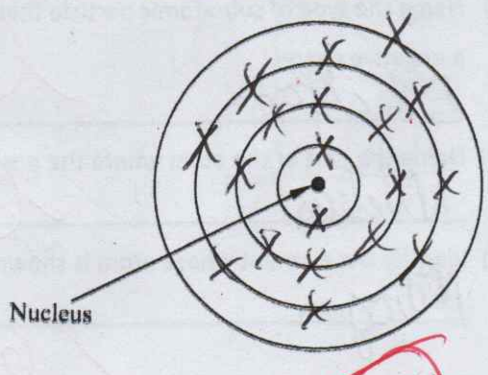
(*iii*) Name an element which has the same number of electrons in its outer shell as fluorine. Chlorine 2,7

3

Question 4

(*d*) A potassium atom has atomic number 19 and a mass number of 39.

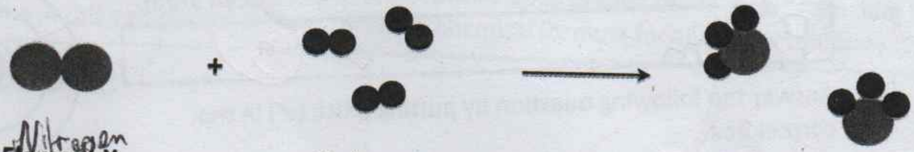
Complete the diagram using dots or crosses to clearly show the arrangement of electrons in the potassium atom.



6

Question 5

(b) Elements consist of atoms. The diagram below represents a chemical reaction between two elements. Element **X** reacts with hydrogen to form compound **Y**.



Nitrogen
Element X | **Hydrogen** | **Compound Y**

(i) Identify element **X** by putting a tick (✓) in the correct box below. (You may use page 79 of the *Formulae and Tables* booklet to help you answer this question.)

Lithium ☐ Magnesium ☐ Nitrogen ☑

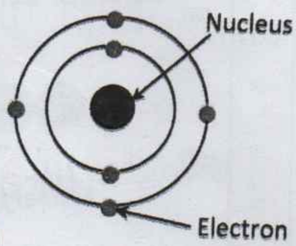
6

(ii) Hence, write the chemical formula for compound **Y**.

H_3N NH_3

Question 6

An atom of element **X** is shown in the diagram.



(a) Name one subatomic particle found in the nucleus of an atom.

Proton

3

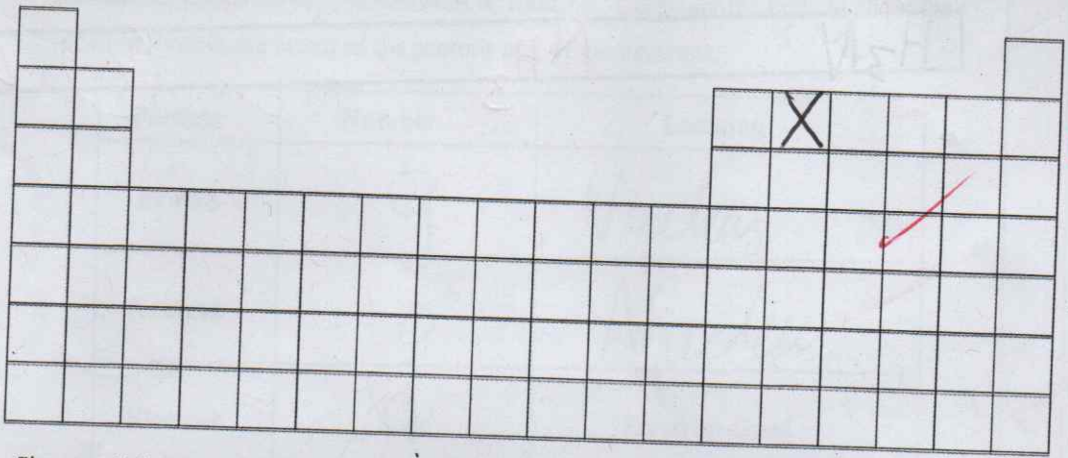
(b) Answer the following question by putting a tick (✓) in the correct box.

The charge on the electron is:

Positive ☐ Negative ☑ Neutral ☐

3

(c) Place an **X** on the Periodic Table shown below to indicate the position of element **X**. You may use the Periodic Table on page 79 of the *Formulae and Tables* booklet to help you answer this question.

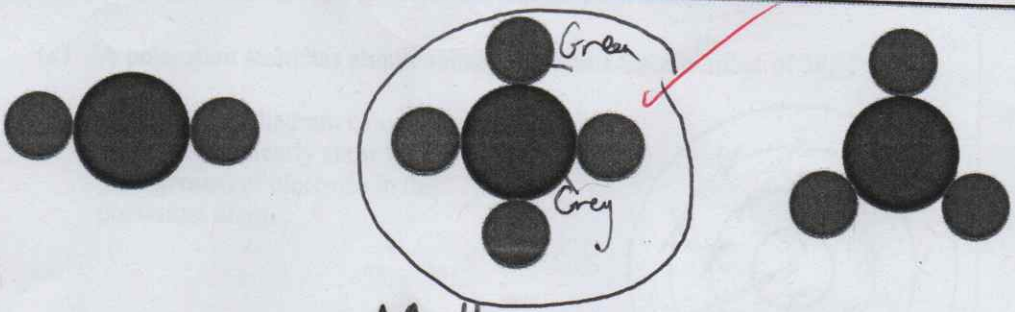


3

(d) Element **X** forms a compound with hydrogen. Element **X** is shown in grey. Hydrogen is shown in green. Circle the diagram below which represents the compound formed. Justify your answer.

Carbon needs 4 electrons to fulfil the Octet Rule

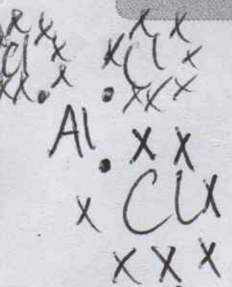
6



Methane

CH_4

Question 7



Rough work

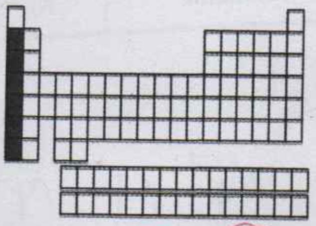
(f) Aluminium reacts with chlorine to form the compound aluminium chloride. Use the Periodic Table on page 79 of the *Formulae and Tables* booklet to predict the ratio of aluminium to chlorine in this compound. Hence write the chemical formula for aluminium chloride.

$AlCl_3$

12

Question 8

(*a*) The diagram is an outline periodic table. One area, a group of elements, is shaded.



Name this group of elements and give one chemical property that they have in common.

Group Alkali Metals

Property 1 electron on outer shell

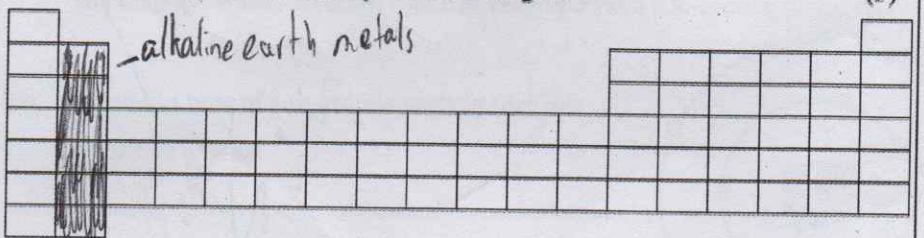
6

Question 9

(*b*) (*i*) Show, clearly using shading and labelling, the ***location*** of the ***alkaline earth metals*** on the blank periodic table given. (3)

For examiner use only

(1) (2)



(*ii*) Name an ***alkaline earth metal***. **Name** Magnesium (3)

6

Question 10

(b) Complete the table below, using the Periodic Table of the elements to predict the ratio of atoms and the chemical formula for each of the compounds listed.

You should refer to page 79 of the *Formulae and Tables* booklet when answering this question.

The first row is completed for you.

I'm not sure

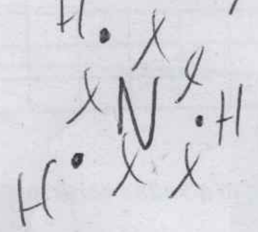
Compound	First element	Second element	Ratio	Formula
Water	Hydrogen (H)	Oxygen (O)	2 : 1	H_2O
Magnesium chloride	Magnesium (Mg)	Chlorine (Cl)	1 : 2	$MgCl_2$
Ammonia	Nitrogen (N)	Hydrogen (H)	1 : 3	NH_3

12

12

Mg^{2+} Cl^- Cl^-

Ionic



Covalent